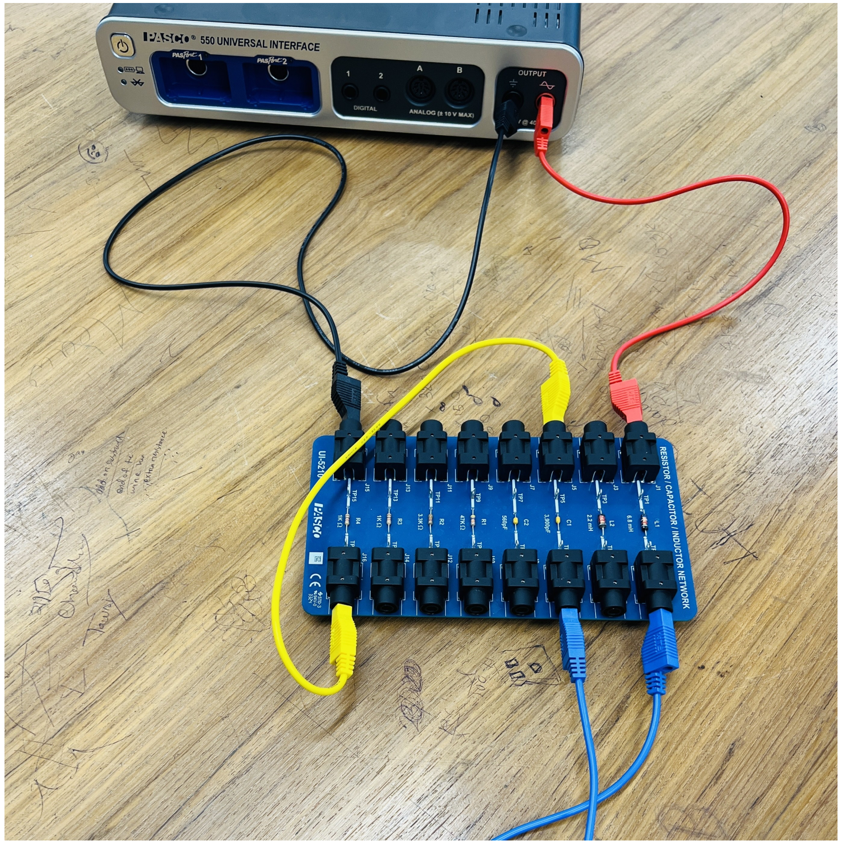
LRC Circuit

Introduction

The current through a series LRC circuit is examined as a function of applied frequency and the effects of changing the values of the resistance, inductance, and capacitance are observed.

The phase difference between the applied voltage and the current is measured below resonance, at resonance, and above resonance.



LRC Circuit

Theory

An inductor, a capacitor, and a resistor are connected in series with a sine wave generator. Since it is a series circuit, the current will be common to all the components a given by

$$I = I_{max} \cos(\omega t) \qquad (1)$$

The voltage across the resistor will be in phase with the current, but the voltage across the inductor leads the current by 90º and the voltage across the capacitor lags by 90º. Adding the three EMF voltages $\mathcal{E}$, results in a total voltage that varies sinusoidally, but ha a phase shift φ with respect to the current and is given by

$$\mathcal{E} = \mathcal{E}_{max} \cos(\omega t + \varphi) \qquad (2)$$

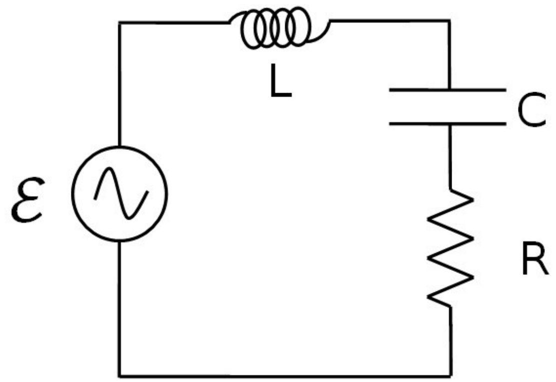
The three components obey the AC analogs of Ohm's Law: $V_R = IR$, $V_{L\,max} = I_{max} X_L$, $V_{C\,max} = I_{max} X_C$, where X_L and X_C are the AC analogs of resistance called the inductive reactance and the capacitive reactance. The maximum current and total voltage are then related by

$$\mathcal{E}_{max} = I_{max} Z \qquad (3)$$

where Z is called the impedance and is the AC analog of resistance for the entire circuit. The phase shift is related to the other variables by

$$\tan \varphi = (X_L - X_C)/R \qquad (4)$$

Since the voltage across the resistor is in phase with the current, the phase of the currer can be measured by measuring the phase of the voltage across the resistor.



Theory (continued)

The impedance Z is given by

$$Z = \sqrt{R^2 + (X_C - X_L)^2} \qquad (5)$$

where the capacitive reactance is

$$X_C = \frac{1}{\omega C} \qquad (6)$$

and the inductive reactance is

$$X_L = \omega L \qquad (7)$$

At resonance, the current is maximum and thus the impedance is at its minimum. The minimum impedance (equation 5) occurs when $X_L = X_C$ yielding Z = R. Setting equation 6 equal to equation 7 yields the resonant frequency:

$$\omega_{res} L = \frac{1}{\omega_{res} C} \qquad (8)$$

$$\omega_{res} = \frac{1}{\sqrt{LC}} \qquad (9)$$

Equipment

1	550 Universal Intrface	UI-5001
1	Resistor/Capacitor/Inductor Network	UI-5210
2	Voltage Sensor	UI-5100
1	Patch Cords (Set of 5)	SE-7123

Setup

1. Connect the Patch cords to the Signal Generator and connect the red cord to one end of the 6.8 mH inductor on the circuit board. Connect the other end of the inductor to the 3900 pF capacitor in series and the 1.0 kΩ in series. Then connect the black cord to the open end of the resistor.

2. Connect a Voltage Sensor to Channel A on the 550 interface and attach the leads across the resistor, making sure the black cable from the voltage sensor is connected to the grounded side of the resistor.

3. Connect a Voltage Sensor to Channel B on the 550 interface and attach the leads across the leads of the Output cable, making sure the black cable from the voltage sensor is connected to the black side of the signal generator.

4. Open the Signal Generator 550 Output and choose the Sine Wave at a frequency of 10 KHz and an amplitude of 7 V. Leave the output on AUTO.

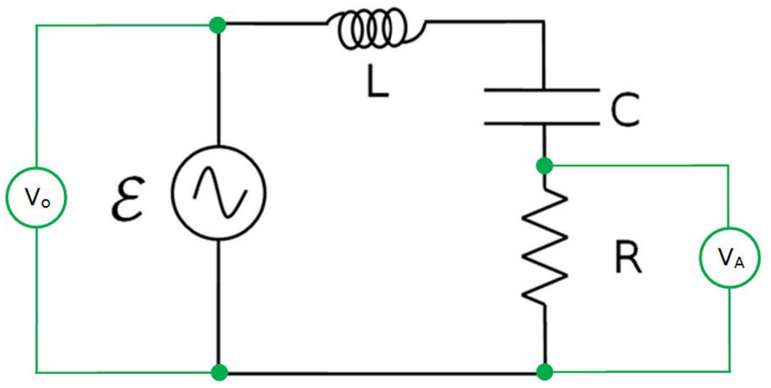


Figure 1: Series LCR Circuit

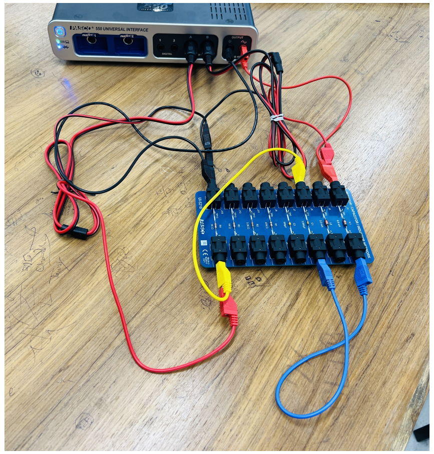


Figure 2 :LRC Circuit with Voltage Sensor

Procedure: Plotting the Resonance Curve

In this part of the lab, you will vary the frequency of the applied voltage and record the response (current) of the circuit. The response is measured by measuring the voltage acr the resistor since the current is in phase with this voltage and it is proportional to it.

One further complication is that you must divide the resistance voltage by the output (of the 550) voltage to account for any changes in the output voltage. This works because the output voltage doubles, then the resistance voltage also doubles and the ratio V_R/Vo remains constant. This is faster than trying to adjust the output voltage each time to keep it constant.

Perform the following steps on the next page:

1. Begin with the signal generator set on 5 kHz. Record this frequency in Table I. Click on Monitor. If the trace is rolling left or right, click the trigger on the oscilloscope. Adjust th horizontal scale on the scope so about three cycles show.

2. Stop monitoring and use the coordinates tool to measure the amplitude of each of the voltages and type the results in Table I. The coordinate tool should show three significa figures for voltage, the snap to pixel distance should be 1 (snap disabled) and the delta tool should be on. If not, right click on the tool and change tool properties. Increase the vertical scale to make the signal (especially V_R) you are measuring as large as possible. Always measure the positive peak. Set the coordinate tool so the horizontal line is tanger to the peaks. Leave the vertical scale expanded until after step 3. That will make it easier to measure the phase shift.

3. Increase the frequency of the output by 5 kHz and repeat the measurements.

4. Continue to increase the frequency in steps of 5 kHz up to 100 kHz.

5. To find the resonance peak more precisely, we need more data near the peak. We want readings at 2 kHz intervals for the region within 10 kHz of the peak. For example, if th peak is at 140 kHz, we would like to add points at: 136 kHz, 138 kHz, 142 kHz, & 144 kHz. To do this, on the Table 1, click and drag to highlight the 130 kHz row. From the grap toolbar, click on insert empty cells above. Then add a run at 125 kHz. Repeat for each desired data point.

7. Exchange the 1 kΩ resistor for the 3.3 kΩ resistor and repeat the entire procedure. In Table 1, click each of the headers (row 1) and select 3.3 k Ohm instead of 1 k Ohm.

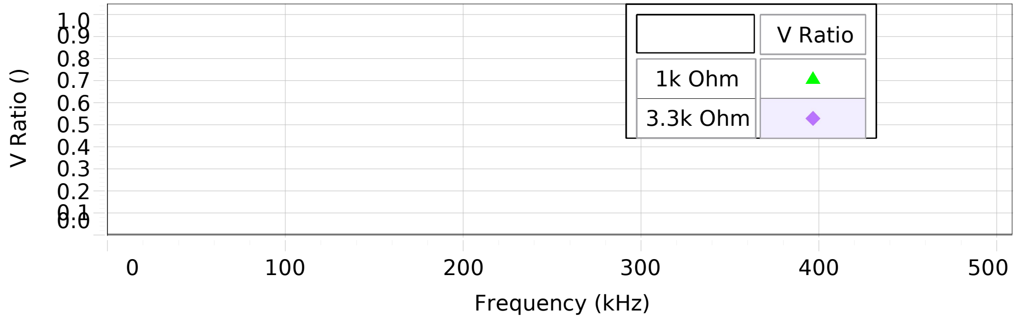


Figure 3: Resonance Curve

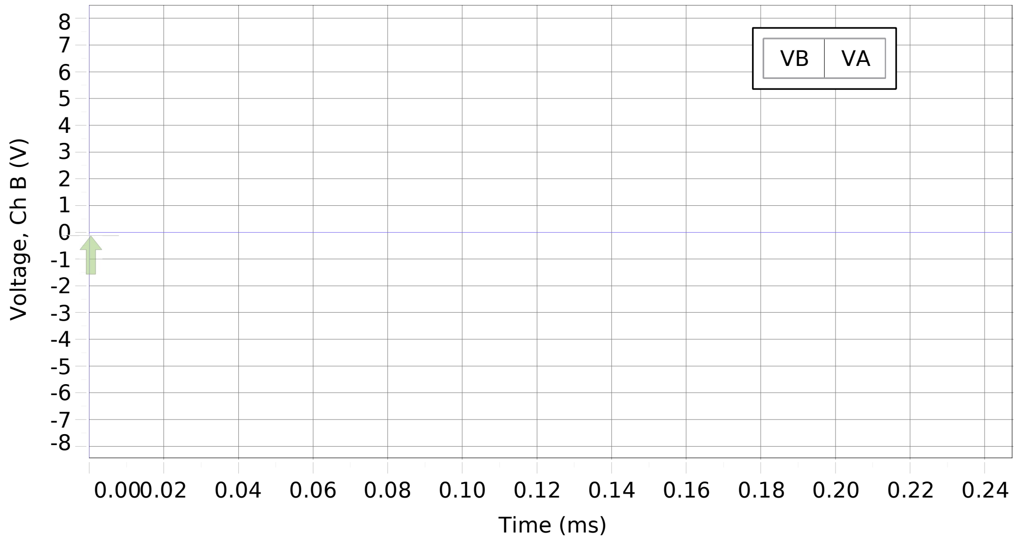


Figure 4: Oscilloscope with Voltages Across Resistor and Output

Table I: Data for Resonance Curve

1k Ohm	1k Ohm	1k Ohm	1k Ohm
Frequency (kHz)	Output VB (V)	Resistor VA (V)	V Ratio

Analysis of the Resonance Curve

1. Display the runs for both of the resistors on the graph.

2. Measure the height of the peak for each curve and the frequency of each peak. You might want to expand the horizontal scale. Record them in the box below.

3. Measure the width of the curve at half the max height for each resistor. As before, you should set the snap to pixel distance to 1 for the coordinate tool.

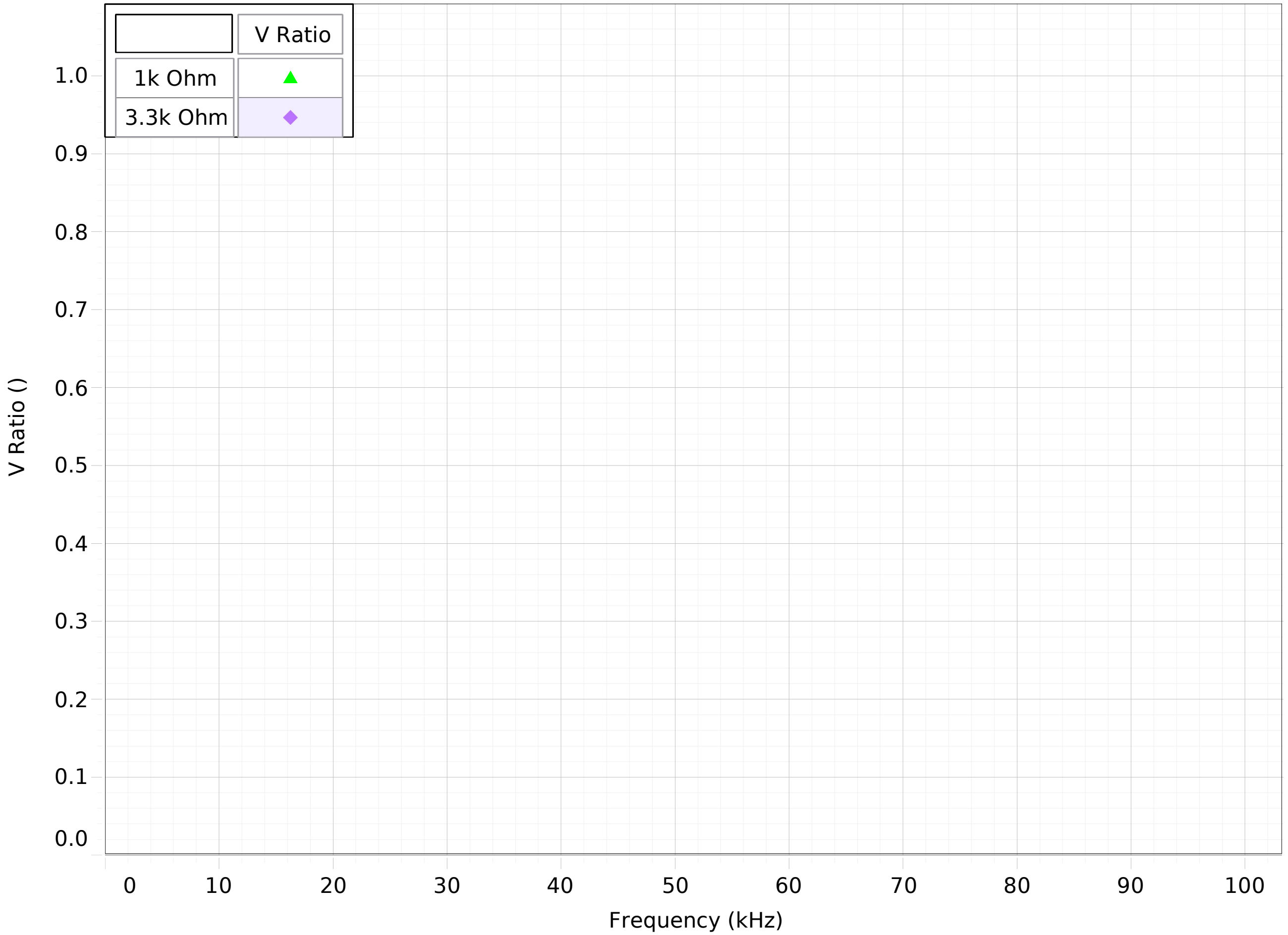


Figure 3: Resonance Curve

Questions

1. How do the height and width of the curves change when you increase the resistance?

2. Calculate the theoretical resonant frequencies and compare them to the measured values with a percent difference. Remember that the frequency of the signal generator is f which is related to the theoretical frequency by

$$f_{res} = \frac{\omega_{res}}{2\pi}$$

$$\omega_{res} = \frac{1}{\sqrt{LC}}$$

3. Does the resistance change the resonant frequency? How does the resistance of the inductor affect the results?

4. Why don't the peaks equal one at the resonant frequency?

5. Why isn't the resonant curve symmetrical about the resonant frequency?

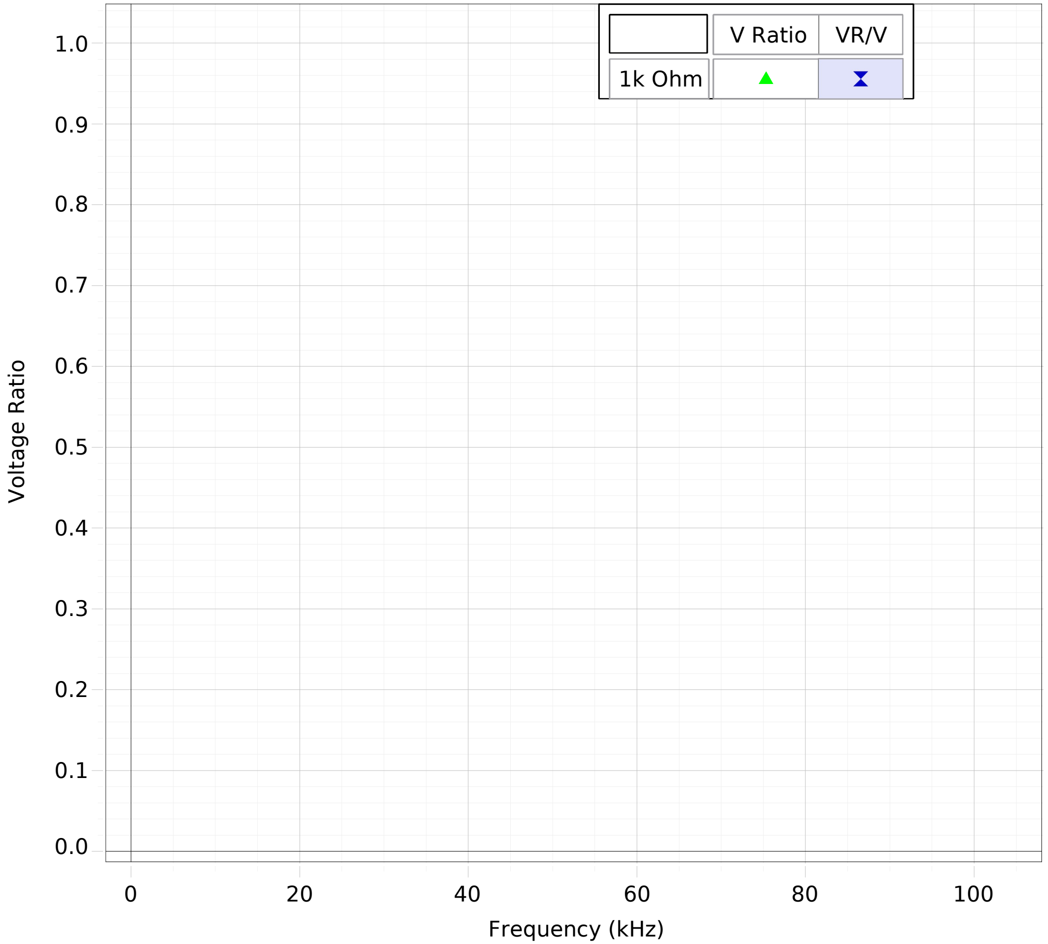


Figure 6: Resonance Curve Model

Computer Model:

1. Click on line 9 in the calculator and verify that the equation corresponds to the voltage ratio, $(V_R/V) = R/Z$, using Equation (5), except that the total resistance is R+r, where r is the resistance of the coil.

2. Vary the values of r, L, & C to fit the curve as well as possible for frequencies **below 100 kHz** (see 3. below). Answer the following questions:
   a. What does r affect?
   b. What does C affect?
   c. What does L affect?

3. Once we have a functioning model we can use it to examine the system. For example, instead of doing the 3300 Ohm experiment, we could just use the model. Try it by changing R to 3300 Ohm. Wow, that's a lot easier!